

Regression inference and robust standard errors

Slide route: Lecture4v2.tex: estimation and hypothesis testing, Stata application, heteroskedasticity, Gauss–Markov. Chapter 5.

Session 5 | Read a regression responsibly

Outcomes: Read an estimate and SE, test a slope, construct an interval, explain robust SEs, and state what Gauss–Markov does and does not guarantee.

100-minute route

0–10	Retrieve OLS versus causality.
10–30	Hypothesis Testing and a regression output table.
30–45	Stata Application: estimate, SE, test, interpretation.
45–55	Residual plots and pause.
55–75	Homoskedasticity versus heteroskedasticity; robust SEs.
75–90	Gauss–Markov, limits, and Midterm 1 synthesis.
90–100	Exit check and explain proposed exam coverage.

Slide-by-slide script

Slide 1 | Title

Guide Frame the class as learning to read regression output without confusing precision with causality.

Slide 2 | Roadmap

Guide Tell students the route: coefficient, test, application, heteroskedasticity, robust inference, and Gauss–Markov.

Slide 3 | Last lesson

Guide Retrieve the OLS slope and conditional mean zero. Ask which assumption is about interpretation rather than arithmetic.

Slide 4 | Last lesson

Guide Use the three-point regression from the previous class to identify a fitted value and a residual.

Slide 5 | Estimation of the regression line

Guide Remind students that the estimate is a random variable before sampling and a number after observing data.

Slide 6 | Estimation of the regression line

Guide Ask what a standard error measures: sampling uncertainty under a specified variance calculation.

Slide 7 | Hypothesis testing

Guide Write $H_0 : \beta_1 = 0$ and $H_1 : \beta_1 \neq 0$. Keep the coefficient's units in the sentence.

Slide 8 | Hypothesis testing

Guide Compute $t = (\hat{\beta}_1 - 0)/SE(\hat{\beta}_1)$ from a displayed estimate. Ask students to identify the numerator.

Slide 9 | Stata application

Guide Read one output row left to right: coefficient, standard error, test statistic, and p-value. Do not read every row.

Slide 10 | Stata application

Guide Use $\hat{\beta}_1 = -2.3$ and $SE = .7$ to calculate $t = -3.286$. Ask for the 95 percent interval.

Slide 11 | Stata application

Guide Interpret the interval and p-value together. A significant slope is not automatically a causal effect.

Slide 12 | Stata application

Guide Ask what sample, outcome, and regressor the command uses. Reproducibility starts with knowing the data behind the table.

Slide 13 | Stata application

Guide Contrast statistical and economic significance using the size of the class-size coefficient.

Slide 14 | Stata application

Guide Show that changing the confidence level changes the critical value, not the point estimate.

Slide 15 | Stata application

Guide Use the output to formulate one complete result sentence with sign, units, uncertainty, and design caveat.

Slide 16 | Stata application

Guide Ask students to diagnose whether a difference between two estimates is sampling noise or a change in specification.

Slide 17 | Homoskedasticity versus heteroskedasticity

Guide Draw constant-width and fan-shaped residual clouds. Define the issue as a variance condition, not a mean condition.

Slide 18 | Homoskedasticity versus heteroskedasticity

Guide Explain that OLS can still estimate the conditional mean slope while conventional SEs fail under heteroskedasticity.

Slide 19 | Graphical illustration

Guide Have students predict where the residual variance is largest. Link the picture to why robust SEs can matter.

Slide 20 | Graphical illustration

Guide Ask whether robust SEs change fitted coefficients. No: they change the uncertainty calculation.

Slide 21 | Robust standard errors

Guide Return to the synthetic example: conventional $SE = .7$, robust $SE = 1.2$. Recompute the interval.

Slide 22 | Robust standard errors

Guide Stress that robust does not mean immune to serial correlation, clustering, selection, or endogeneity.

Slide 23 | In practice

Guide Tell students to report which covariance estimator was used and why it matches the sampling structure.

Slide 24 | In practice

Guide Ask them to identify the observational unit and the possible dependence clusters before choosing standard errors.

Slide 25 | Theoretical foundations

Guide State the conditions behind unbiasedness and precision without presenting them as automatic facts.

Slide 26 | Gauss–Markov

Guide Explain BLUE as best linear unbiased, under homoskedasticity and the other maintained assumptions.

Slide 27 | Gauss–Markov limits

Guide Say clearly what the theorem does not provide: causal identification, protection from omitted variables, or best performance among all nonlinear estimators.

Slide 28 | Back to the original question

Guide Return to the class-size example and ask what would still be needed for a causal claim.

Worked example | One coefficient, two SEs

Use with slides Lecture4v2 slides 7–11 and 17–22. Use the coefficient test during Hypothesis Testing/Stata Application, then return to the same estimate when the deck reaches heteroskedasticity and robust standard errors.

Synthetic regression: test score on class size, $\hat{\beta}_1 = -2.3$. Suppose conventional $SE = .7$ and heteroskedasticity-robust $SE = 1.2$.

$$t_{\text{conventional}} = -3.286, \quad t_{\text{robust}} = -1.917.$$

Large-sample 95% intervals are

$$-2.3 \pm 1.96(.7) = [-3.672, -.928],$$

$$-2.3 \pm 1.96(1.2) = [-4.652, .052].$$

The estimated association is identical. The uncertainty assessment changes; at the 5% two-sided level the robust calculation fails to reject zero.

Important qualification Robust SEs are not always larger. This example illustrates a possible difference, not a general inequality. With independent observations, heteroskedasticity-robust inference tolerates unequal conditional error variances; it does not automatically handle serial or cluster dependence.

Board intuition and classroom implementation

Use with slides Lecture4v2 slides 17–22 for the residual-spread drawing and robust-SE formula; slides 23–25 for the Gauss–Markov interpretation and its limits.

Draw two residual plots: constant vertical spread and a widening spread. Both can have conditional mean zero. Heteroskedasticity concerns $\text{Var}(u | X)$, not necessarily $\mathbb{E}(u | X)$.

$$\widehat{\text{Var}}_{HC0}(\hat{\beta}_1) = \frac{\sum_i (X_i - \bar{X})^2 \hat{u}_i^2}{\left[\sum_i (X_i - \bar{X})^2\right]^2}.$$

Explain the weighting intuition rather than deriving the entire sandwich matrix. Software may use a finite-sample adjustment, such as HC1, instead of HC0.

Illustrative Stata commands For a prepared dataset with variables named `score` and `classsize`:

```
reg score classsize, vce(robust)
test classsize = 0
```

These names are placeholders for a classroom example; check the actual dataset's variable names before the class. If no dataset is available, use the synthetic output on page 1.

Gauss–Markov in words With the classical linear-model assumptions, including homoskedasticity and appropriate exogeneity, OLS is best among linear unbiased estimators. This is not a theorem saying every regression is causal, or that OLS is best among all estimators.

Prompts and answers

Does robust mean unbiased? No. A robust SE can accompany a biased coefficient when regressors are endogenous.

Would more observations cure omitted-variable bias? No. They can make an estimate of the wrong target very precise.

What does an interval spanning zero mean? The data do not reject a zero slope at that corresponding level; many nonzero slopes also remain compatible.

Midterm 1 preparation Proposed coverage: Lecture 1 probability/sampling, Lecture 2 mean inference, Lecture 3 OLS, Lecture 4 regression inference. Students should practice a joint table, a mean test, a hand-computed OLS line, and a robust-SE interpretation.

If short Reduce repeated Stata screenshots. Preserve coefficient/SE interpretation and the distinction between heteroskedasticity and endogeneity.

Source: Lecture4v2.tex. Numerical regression output is illustrative.